

Modeling the Spread of Wildfires and Assessing California's Urban Energy Infrastructure Vulnerability

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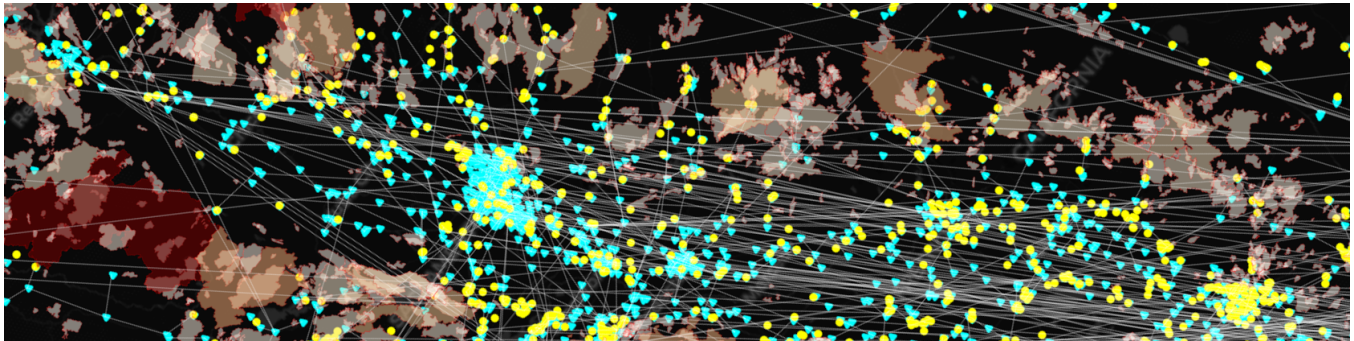


Figure 1: A Magnified Map of California Wildfires and Energy Infrastructure.

Abstract

Wildfires, driven by increasing global temperatures and climate change, present a growing threat to urban communities, especially by posing a risk to energy infrastructure that serves millions of consumers. California, in particular, faces heightened vulnerability with high urban populations, extensive energy infrastructure, and frequent wildfire activity. In this paper we seek to present a framework to examine wildfire spread and its impact on critical Californian energy systems. By leveraging historical wildfire area data in conjunction with spatial data of substations, power plants, and transmission lines, we construct and overlay graph-based models for further analysis. Through clustering algorithms and network metrics, we identify infrastructure components most susceptible to disruption via wildfires and quantify risks to these grid nodes. Combining these perspectives provides insights to aid in resilience planning to inform strategies for safeguarding crucial elements of California's energy network, ensuring more reliable power access for the state's millions of residents.

Keywords

Wildfires, California, Energy infrastructure, Clustering algorithms, Network analysis, DBSCAN, Urban computing

1 Introduction

Wildfires have become increasingly frequent and severe across the United States due to rising global temperatures, prolonged droughts, and shifting climate patterns. These fires present a growing threat not only to natural ecosystems but also to urban communities, where dense populations and critical infrastructure increase the risks. Since the 1980s, the average number of acres burned annually has nearly tripled [2], highlighting the urgent need for research into both wildfire behavior and strategies to mitigate their potentially disastrous effects.

California, in particular, faces heightened vulnerability. The state's combination of large urban populations, extensive energy infrastructure, and frequent wildfire activity creates a unique challenge for resilience planning. Energy systems, such as substations, power plants, and transmission lines, are essential to sustaining daily life and economic activity, yet they are increasingly exposed to wildfire hazards. Disruptions to these systems can cause widespread power outages, threatening millions of residents and critical services.

In this paper, we present a framework to examine wildfire spread and its impact on California's energy infrastructure. By leveraging historical wildfire area data alongside spatial datasets of grid components, we construct graph-based models that allow for systematic analysis of vulnerability. Through clustering algorithms and network metrics, we identify infrastructure nodes that are most susceptible to wildfire disruption and quantify risks to these critical grid elements through a novel metric for resilience planning purposes with future infrastructure development.

Our findings show that clusters of energy infrastructure near constant wildfire regions exhibit higher composite risk scores, with this score computed by direct exposure to the fire, regional fire levels, as well as betweenness and outage simulations. These insights

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highlight that resilience planning measures must account not only for past fire impacts but also for regional spatial patterns of recurring wildfire activity, enabling policymakers to prioritize hardening and mitigation strategies in areas most likely to experience future wildfire damage.

Ultimately, this work contributes to research into climate resilience by providing actionable strategies for safeguarding California's energy systems and a metric for quantifying the risk at a given location. By integrating wildfire modeling with infrastructure vulnerability analysis, we aim to support more reliable power access for millions of residents in the face of escalating climate-driven wildfires.

2 Related Research

Wildfire behavior has been recognized as the outcome of a complex interplay between fuel availability, weather conditions, and climate. Traditional methods provide important insights into these systems, but fall short of capturing the complex non-linear processes that drive the spread of wildfires [5]. Recent studies have shown using machine learning techniques to estimate the importance of these environmental factors proves to be useful in predicting where future fires might start and how climate change and wildfires are connected.

One article explains how support vector machines and ensemble methods have shown promise with environmental datasets, but seem to struggle with the dynamic nature of wildfires [1]. Deep learning approaches, such as convolutional neural networks and convolutional recurrent networks, demonstrated the ability to capture spatio-temporal complexities of wildfire data that the previous models had missed.

At the same time, we see published articles bringing attention to the importance of finding vulnerabilities in urban energy infrastructure. Another paper describes the vulnerabilities in urban energy infrastructure through three stages in the novel COVID analogy [3]. Stage 1 (virus entry) describes an extreme event disrupting a component of the energy ecosystem. Stage 2 (propagation through respiratory system) describes an electricity sector failing in a whole area. Stage 3 (heart, kidneys, digestive system, and other organs) describes penetration beyond the energy ecosystem and to other interconnected systems such as transportation, water supply, and communication. This explains why building a resilient energy infrastructure is pivotal for the success of an urban system.

A paper assessing the risk to California's energy infrastructure from projected climate change highlights how increasing wildfire frequency and severity directly threaten transmission corridors, substations, and generation facilities [4], with it being noted that wildfires have a significant effect on transmission lines, besides just affecting the substations. With the soot of the wildfires accumulating on these lines, wildfires can cause great damage even by being in the vicinity of key transmission lines, causing downstream impacts.

3 Methodology

3.1 Isolated Analysis of Wildfires

To characterize wildfire activity in California, we constructed a graph representation of the MTBS Burned Area Boundaries dataset.

Each wildfire event was represented as a node, with attributes including burned area size and severity. Edges were introduced between nodes when the corresponding fire perimeters spatially overlapped, thereby capturing relationships among fires that affected similar regions in order to locate key hotspots.

The shapefiles were first filtered to retain only events associated with California. Data were reprojected to EPSG:3310 (NAD83/California Albers) for accurate spatial operations, and subsequently transformed to EPSG:3857 (Web Mercator) for visualization. Centroids of each fire polygon were computed to facilitate plotting and edge representation.

The resulting wildfire graph was analyzed using standard network measures. Degree centrality quantified the number of overlaps per fire, clustering coefficients measured local connectivity, and betweenness centrality identified fires that made connections between clusters. Connected components were extracted to identify groups of fires with overlapping spatial footprints. These metrics provided insight into the structural properties of wildfire occurrence and overlap, regardless of energy infrastructure for now.

3.2 Clustering of Wildfires

To identify recurrent fire-prone zones in California, we applied clustering techniques to the MTBS wildfire perimeter dataset. Each fire perimeter was represented by its centroid after reprojection to the California Albers coordinate system (EPSG:3310), ensuring accurate distance calculations. This preprocessing step allowed for a simpler geometric system focused on California, ensuring more accurate computation of distances in this region. As seen below, these reprojections are constantly updated based on the specific use with data coming from the shapefile as EPSG:4326, EPSG:3310 for regular computation, and EPSG:3857 for basemap display.

Algorithm 1 Wildfire Clustering with DBSCAN

Require: Wildfire perimeters P , DBSCAN parameters ϵ , minPts

Ensure: Cluster labels for each fire perimeter

- 1: Reproject P from EPSG:4326 to EPSG:3310
 - 2: Compute centroids C from geometries in P
 - 3: Apply DBSCAN on C with parameters $(\epsilon, \text{minPts})$
 - 4: Obtain cluster labels L for each centroid
 - 5: Attach labels L back to corresponding perimeters in P
 - 6: Reproject P to EPSG:3857 for basemap display
 - 7: Plot perimeters colored by cluster label
 - 8: Add legend and basemap tiles
-

We used the DBSCAN algorithm outlined in Algorithm 1 to group fire events into clusters based on spatial proximity. This algorithm was chosen because it can discover irregularly shaped clusters without requiring the number of clusters to be specified in advance, and it naturally distinguishes between dense regions of fire where wildfires are frequent and more isolated events that are just noise. Parameters were tuned to reflect realistic fire spread distances: the neighborhood radius was set to 20 km and the minimum number of fires to form a cluster was set to 17. These values balanced the trade-off between overly fragmented clusters and a

single widespread cluster. Noisy data was explicitly labeled as cluster -1 and visualized in light gray to highlight their distinction from persistent fire regions.

In addition to DBSCAN, we experimented with Kernel Density Estimation (KDE) to generate continuous hotspot surfaces for a smoother intensity map of fire centroids, reinforcing patterns visible in the raw overlay. While KDE highlights general density trends, DBSCAN produces discrete and analyzable clusters that can be directly connected with the energy infrastructure layers in the upcoming sections. The resulting clusters serve as the foundation for risk analysis. By combining hundreds of individual fire events into meaningful spatial groupings, clustering enables us to quantify frequency and proximity to critical infrastructure such as transmission lines and substations. This representation of wildfire activity is essential for computing risk metrics and identifying resilience measures as needed.

3.3 Isolated Analysis of Energy Infrastructure

To study the energy infrastructure in California, we created an undirected graph. The graph was constructed from two datasets (a substation dataset with geographic coordinates and facility meta-data and a transmission line dataset with connections between substations). Additionally, we stored voltage as an edge attribute used for future metrics. The graph is undirected since directionality was not encoded in our dataset for transmission lines. We characterized substations as nodes and transmission lines as edges. We preprocessed this data by removing duplicate assets, dropping rows with missing substation names, and ensured consistent typing.

We also compute a few standard metrics to begin learning about our graph.

- Degree Centrality - counting direct connections to other substations.
- Betweenness Centrality - measures importance in a network by checking how often it lies in the shortest path of two substations.
- Clustering Coefficient - degree which substations in the graph tend to cluster together.
- Connected Components - number of subgraphs where each substation is fully connected to one another.

After graph construction, we perform basic validation checks, such as node/edge counts and isolated node count to ensure our network was properly created.

3.4 Clustering of Energy Infrastructure

To properly capture how nearby facilities might be damaged during a wildfire, we perform spatial clustering of the energy infrastructure. In other words, we identify nearby groups of substation infrastructure rather than treating substations individually. Again, we choose the DBSCAN algorithm to perform this task as it can identify clusters of any shape while also not requiring numbers of clusters in advance. California's energy grid is not evenly spread, with cities having denser clusters of substations while rural areas are more sparse. DBSCAN is also proficient at handling outliers, meaning the isolated rural substations are left unclustered rather than being forcibly added to unreasonably far facilities (like k-means would).

DBSCAN requires us to choose our ϵ (eps) and Minimum Samples (min_samples). The epsilon value determines the maximum distance between nodes to be considered neighbors (in our case, 0.05 degrees). However, this required some tuning as some clusters ended up being too large. Minimum Samples determines the minimum number of points required to create a dense region, which we have set to 3.

Algorithm 2 Energy Infrastructure Clustering with DBSCAN

Require: Substation graph G , DBSCAN parameters ϵ , minPts

Ensure: Cluster labels for each substation

- 1: Extract coordinates for each substation (node) in G
 - 2: Apply DBSCAN on coordinates with parameters $(\epsilon, \text{minPts})$
 - 3: Obtain cluster labels L for each substation
 - 4: Label substations with fewer than minPts as noise
 - 5: Reproject substations to EPSG:3310 for spatial analysis
 - 6: Reproject to EPSG:3857 for basemap display
 - 7: Plot substations with cluster labels in legend
-

3.5 Developing a Risk Metric

To evaluate the vulnerability of clustered substations to wildfire exposure and network fragmentation, we construct a composite risk metric that integrates three components: outage impact, network centrality, and wildfire exposure. Each component is normalized to the unit interval $[0, 1]$ to ensure proper comparability and be able to be more readily understandable.

The first step in this process is clustering substations using DBSCAN as described in the previous subsection with noise being assigned a cluster ID of -1 . With these clusters, we create a GeoDataFrame consisting of rows for each cluster with metrics for each of these clusters. While plants were used for general analysis, they are excluded from this metric for reasons that will be covered in depth in the 'Limitations' section, but due to no transmission line data for these plants and far too many possible inconsistencies to make any assumptions, it was safer to remove plants when considering this risk score.

The key component of this GeoDataFrame construction is the simulation of cluster outages in the event that wildfires happen to take out this particular cluster of substations. See Algorithm 3 below for the pseudocode of the simulation process, but essentially, this allows us to simulate the consequences of losing all substations in a given cluster, and quantify how this affects the overall grid network. After these substations are removed, the connected components of the remaining overall graph are examined to extract "stranded" components (components that have fewer than s_{min} substations) and add this to the number of removed nodes as part of this cluster to get the percent of the network impacted. Additionally, degree centrality metrics are utilized to find how much connectivity is lost when these substations are considered to be failed.

Finally, we begin the computation of the composite risk using the metrics from outages, betweenness, and fires. This is computed by cluster rather than by substation in order to provide a better idea of risk with regards to all possible connections of nodes within these clusters and allow for classification based on general regions, which can be more easily interpretable. To compare clusters on a common

Algorithm 3 SimulateClusterOutage

Require: Infrastructure graph $G = (V, E)$, cluster nodes $C \subseteq V$, minimum viable size $s_{\min}$

Ensure: Dictionary of impact statistics

```

1:  $H_0 \leftarrow \text{copy of } G$ 
2:  $B \leftarrow C \cap V(H_0)$  ▷ burned nodes
3:  $H_1 \leftarrow \text{copy of } H_0$ 
4: remove nodes  $B$  from  $H_1$ 
5:  $stranded \leftarrow 0$ 
6: if  $|V(H_1)| > 0$  then
7:    $stranded \leftarrow \sum_{W \in CC(H_1)} \mathbf{1}\{|W| < s_{\min}\} \cdot |W|$ 
8: end if
9:  $totalImpacted \leftarrow |B| + stranded$ 
10:  $baselineSize \leftarrow |V(H_0)|$ 
11:  $pctImpacted \leftarrow 0$ 
12: if  $baselineSize > 0$  then
13:    $pctImpacted \leftarrow totalImpacted / baselineSize$ 
14: end if
15:  $deg \leftarrow \text{degree dictionary of } H_0$ 
16:  $directDegreeLoss \leftarrow \sum_{v \in B} \deg(v)$ 
17:  $largestCompSize \leftarrow \max\{|W| : W \in CC(H_1)\}$ , default 0
18:  $pctGiant \leftarrow 0$ 
19: if  $baselineSize > 0$  then
20:    $pctGiant \leftarrow largestCompSize / baselineSize$ 
21: end if
22: return  $dict$  ▷ dict with calculated attributes

```

scale, each raw metric is first normalized to the unit interval $[0, 1]$ using min–max normalization. For example, the number of directly destroyed substations in an arbitrary cluster ℓ is normalized as

$$D'_\ell = \frac{D_\ell - \min(D)}{\max(D) - \min(D)} \quad (1)$$

Similar normalization is applied to the stranded count S_ℓ and degree loss L_ℓ .

The outage score O_ℓ is constructed as a weighted combination of these normalized proxies:

$$O_\ell = 0.25 S'_\ell + 0.20 D'_\ell + 0.15 L'_\ell \quad (2)$$

This reflects both direct destruction, stranded nodes, and connectivity loss.

We compute the average betweenness centrality of substations in the cluster and normalize it using Equation 1, with v representing an individual substation node and C_ℓ representing the set of all nodes in the cluster ℓ :

$$B_\ell = \frac{1}{|C_\ell|} \sum_{v \in C_\ell} \text{betweenness}(v) \quad (3)$$

Wildfire risk combines burned area, severity, fire frequency, and proximity to the nearest fire. Proximity is modeled with exponential decay to lessen the overall risk as the substation is further away from any fires:

$$P_\ell = \exp\left(-\frac{d_\ell}{\lambda}\right) \quad (4)$$

where d_ℓ is the distance to the nearest wildfire in kilometers and $\lambda = 10$ is the decay parameter. As previously discussed with wildfire clustering, this information is used to reflect overall regional

effects of wildfires in a particular area. With a Gaussian kernel, the kernel density estimate of the wildfires is also incorporated into the wildfire component of the score with K'_ℓ . While this informs us of the overall density of fire activity, the DBSCAN wildfire clusters tell us where fires repeatedly cluster. The following formula is used to determine the contribution of the nearest DBSCAN cluster of wildfires in the area.

$$D_\ell = \exp\left(-\frac{d_\ell^{\text{cluster}}}{\lambda}\right) (0.5A_\ell^{\text{cluster}'} + 0.3H_\ell^{\text{cluster}'} + 0.2N_\ell^{\text{cluster}'}) \quad (5)$$

Assuming that A_ℓ represents the normalized burned acreage, H_ℓ represents the high severity acreage, and N_ℓ represents the number of fires, the wildfire score is then

$$F_\ell = 0.20A'_\ell + 0.30H'_\ell + 0.35P_\ell + 0.15N'_\ell + 0.10K_\ell + 0.10D_\ell \quad (6)$$

Note that the proximity is not normalized as this is defined by the exponential decay function, so no additional normalization is needed. Upon one further round of weighting for the outage score, the betweenness score, and the wildfire score, these can be summed to output a final composite risk metric, R_ℓ , that assesses the risk for each cluster of substations.

$$R_\ell = 0.35O_\ell + 0.25B_\ell + 0.40F_\ell \quad (7)$$

The weights chosen throughout this process was derived from a determination of which sections have the most value in the overall score.

4 Experiments

4.1 Data

The data for wildfires and energy infrastructure was processed in slightly different ways due to project needs and data formats.

4.1.1 Wildfire Data. We use the MTBS Burned Area Boundaries dataset, which provides polygon shapefiles with the perimeters of wildfire events. Each fire is represented by multiple companion files in the ESRI Shapefile format: .shp (geometry), .shx (index), .dbf (attributes), and .prj (projection). These files were loaded into Python using the GeoPandas library, which preserves both spatial geometry and metadata in a GeoDataFrame. After import, the data were standardized to a common coordinate reference system, with EPSG:3857 being used for plotting, while EPSG:3310, a California-based system, was used for more accurate analysis. This data was then cleaned to remove records that are not associated with the state of California or are incomplete. The perimeter polygons were then used for overlay analysis with energy infrastructure.

4.1.2 Energy Infrastructure Data. Energy infrastructure data were obtained from the Homeland Infrastructure Foundation-Level Data (HIFLD) repository. Three datasets were used: power plants, substations, and transmission lines, each provided in a CSV format. These were processed using Pandas dataframes, with attributes such as geographic coordinates and facility type retained for analysis.

4.2 Experimental Setup

Our experimental setup focuses on validating some of our modeling choices in this project, rather than jumping to a single configuration. Since the methods we have described for spatial clustering and

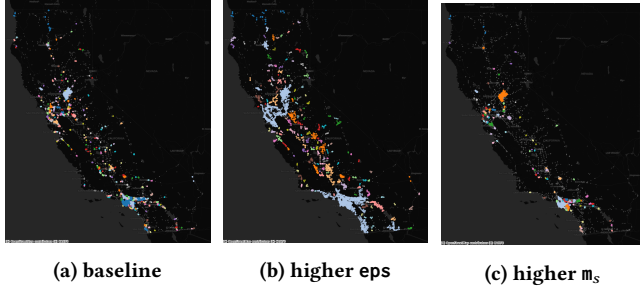


Figure 2: DBSCAN sensitivity analysis. Baseline: $\text{eps}=3000\text{m}$, $\text{min_samples}=3$. Variant (b) increases eps to 8000m . Variant (c) increases min_samples to 5.

created risk metrics introduce new parameters, we explore how these parameters influence our outcomes step by step.

First, we return to DBSCAN, where we have eps and min_samples . Figure 2 illustrates a drastic reduction in number of clusters when we increase min_samples from 3 to 5. Figure 2 also shows the explosive increase of clusters when we go from $\text{eps}=3000\text{m}$ to $\text{eps}=8000\text{m}$. After reviewing multiple similar cases, we decided to stay with $\text{eps}=3000\text{m}$ and $\text{min_samples}=3$ as it struck a nice balance between number and size of clusters without disregarding smaller cities/grids.

Similarly, when testing with wildfire clusters, we tested k -means clustering along with DBSCAN clustering and found that DBSCAN resulted in more natural clusters. Using an ϵ value of 20 kilometers and min_samples of 17 allowed for well-formed clusters that correlate to significant areas of wildfire activity and avoided picking up too much noise that was not as relevant. In addition to just the DBSCAN clustering, we also created maps of hotspots and experimented with several options with this, eventually landing on using a Gaussian kernel with a bandwidth of 20 kilometers.

Another key parameter that required testing was the various metrics of risk we chose. A central challenge was determining which clusters are more critical for calculating risk. Prior to landing on a mix between outage calculations and betweenness scores, we performed some testing. Figure 3 displays how incorporating outage to our risk score impacts various clusters' ranking. In this particular example, we see some of the larger clusters become darker (higher risk ranking) due to having more impact during an outage. With this, we believe it is an important metric to keep during our experimentation.

Another key metric we considered after performing our initial risk assessment in with transmission line voltage weighting. Our motivation for this change was in realizing higher voltage lines are likely more critical to urban environments than lower voltage lines, which may be serving local areas. We tested this through using unweighted edges, where all transmission lines were treated equal, versus voltage weighted lines, where edges are inversely proportional to voltage, V , preferring higher voltage lines.

$$\text{Edge Cost} = \frac{1}{V} \quad (8)$$

Tables 1 and 2 show us the impact adding voltage weighting has on our overall risk score, and in particular, the betweenness score.

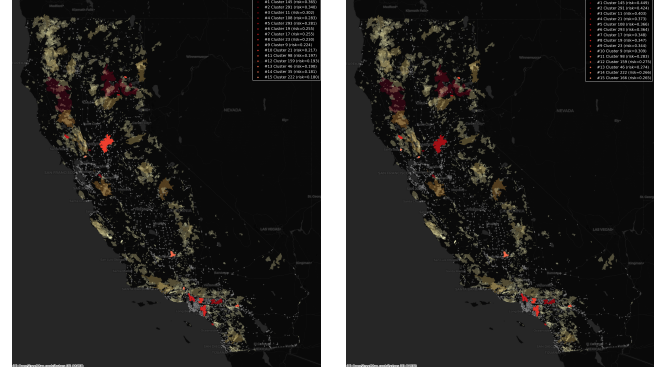


Figure 3: Outage vs. No Outage

Table 1: Top clusters by composite risk (no voltage weighting).

cluster_id	cluster_size	total_impacted	pct_of_network	avg_betweenness	burned_acres	nearest_fire_kn	composite_risk
145	5	1320	0.3115	0.0171	1728657.0000	2.1756	0.4488
291	6	1321	0.3118	0.0135	1583745.0000	0.0000	0.4095
11	80	1382	0.3262	0.0006	3310463.0000	0.0000	0.4064
21	345	1587	0.3746	0.0014	515058.0000	0.0000	0.3815
108	6	1320	0.3115	0.0027	7810186.0000	1.2172	0.3814
17	65	1362	0.3215	0.0018	5477118.0000	0.8751	0.3575
23	98	1408	0.3323	0.0014	3247226.0000	0.0000	0.3566
19	26	1341	0.3165	0.0010	5442754.0000	0.0000	0.3508
293	3	1318	0.3111	0.0098	381098.0000	0.0599	0.3320
97	3	1319	0.3113	0.0119	28241.0000	4.6517	0.3226
9	40	1342	0.3167	0.0005	4331754.0000	1.0668	0.3124
166	5	1322	0.3120	0.0089	887200.0000	1.7787	0.3111
89	4	1321	0.3118	0.0107	44499.0000	4.8998	0.3021
98	7	1323	0.3122	0.0013	4241851.0000	0.4310	0.2902
35	5	1317	0.3108	0.0040	1414150.0000	0.0000	0.2799

Table 2: Top clusters by composite risk (with voltage weighting).

cluster_id	cluster_size	total_impacted	pct_of_network	avg_betweenness	burned_acres	nearest_fire_kn	composite_risk
145	5	1320	0.3115	0.0282	1728657	2.1756	0.4488
291	6	1321	0.3118	0.0239	1583745	0.0000	0.4240
11	80	1382	0.3262	0.0006	3310463	0.0000	0.4026
21	345	1587	0.3746	0.0014	515058	0.0000	0.3726
108	6	1320	0.3115	0.0027	7810186	1.2172	0.3659
293	3	1318	0.3111	0.0197	381098	0.0599	0.3635
17	65	1362	0.3215	0.0018	5477118	0.8751	0.3477
19	26	1341	0.3165	0.0012	5442754	0.0000	0.3465
23	98	1408	0.3323	0.0009	3247226	0.0000	0.3441
9	40	1342	0.3167	0.0005	4331754	1.0668	0.3094
98	7	1323	0.3122	0.0013	4241851	0.4310	0.2827
159	12	1323	0.3122	0.0004	2666874	0.0000	0.2754
46	32	1336	0.3153	0.0002	734223	0.0186	0.2743
222	5	1323	0.3122	0.0091	1858845	5.5491	0.2657
166	5	1322	0.3120	0.0095	887200	1.7787	0.2651

This jump is more noticeable in the two highest risk clusters (cluster_id=145 and cluster_id=291). In terms of overall rank changes, we see the top of the list stay consistent while the bottom parts of the ranking have a few clusters shuffle in risk rank. This might suggest our composite scoring is not dominated by any single metric, however, since we ultimately do not see major changes using this metric, it was removed.

4.3 Results

In our analysis, DBSCAN found 298 distinct clusters across California's energy infrastructure. This encompasses 63.1% of the sub-station network. This also means 36.9% of substations were classified as noise. While 36.9% seems quite large, it is consistent with California's infrastructure. Many of the substations seem to be geographically isolated in more rural areas, and do not form dense

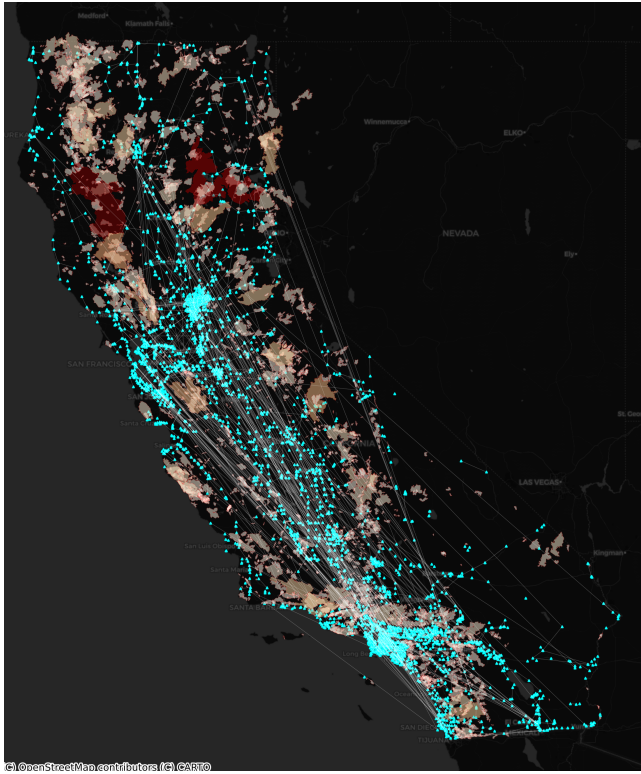


Figure 4: Overlay of California energy infrastructure (blue) with wildfire data (red)

neighborhood as specified by our parameters in DBSCAN. It follows that these substations remain unclustered and not forced into distant groups, which especially makes sense when considering the next steps to overlay the results with wildfire data.

Figure 4 overlays our California energy infrastructure graph data with the wildfire data. It provides a visual reference to ensure both datasets align correctly spatially. This lets us see where wildfire and infrastructure data overlap and separated. We immediately notice a mismatch between the densest areas of the substation network (blue) and wildfire clusters (red). The substations are concentrated in areas with higher population, where the densest areas include the Bay Area and Los Angeles Metropolitan Area. In the wildfire layer, we notice the darkest shaded areas appear in Northern California/mountainous regions, where substations are often more sparse.

A key idea that is displayed when interfacing the two graphs is how close some wildfires borders approach some of the denser clusters of energy infrastructure. Wildland-urban interface (WUI) areas are created by urban planners to prevent wildfire spread in urban areas. Despite this, wildfire spread to nearby areas still often cause grid disruption. The visualization supports the rest of this project’s research as we are motivated to capture information through the overlay of our datasets.

As discussed previously a KDE plot and DBSCAN clustering was developed for the wildfires as seen in Figure 6 with the parameters discussed previously, as determined through testing. With these

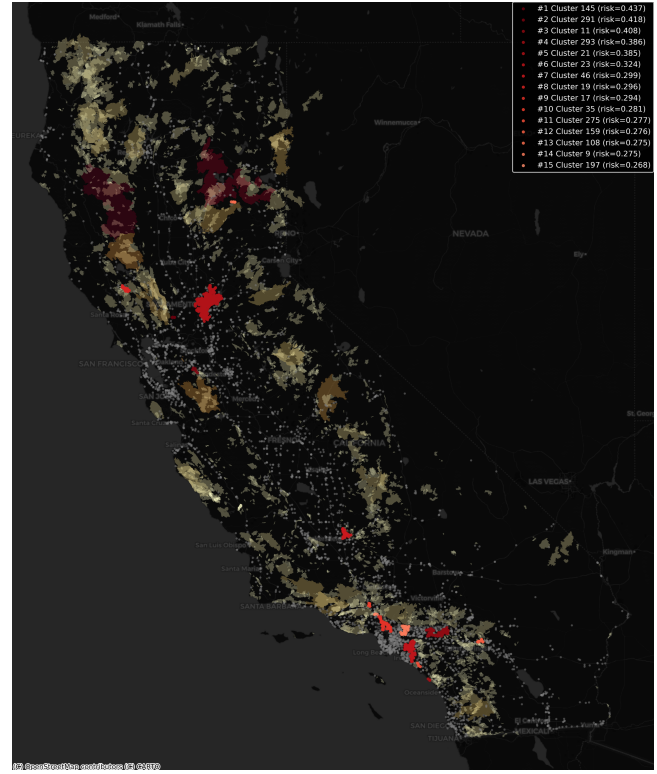


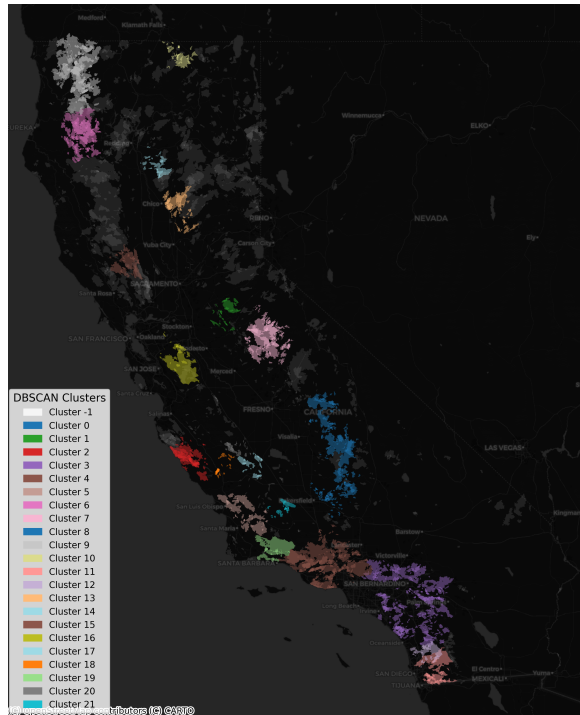
Figure 5: Composite risk map

mapped, we are provided with a better idea of regional wildfire trends, constituting a component of the fire part of the overall risk computation for local context. We can see that most of the very light gray that is classified as noise by the DBSCAN algorithm appears more faint in the KDE plot as well, suggesting that these areas may have occasional fires, but do not see the heavy activity that other regions might, especially along the southern coast of the state.

With the overall risk score computed and mapped, we produce an output as seen in Figure 5 which shows the overall risk scores that loosely correlate to a few of the key hotspots of high risk (labeled in red) of urban infrastructure. In Southern California we notice clusters of high risk alongside edges of wildfire action and dense infrastructure regions. This matches the idea that wildfires do not need to occur within cities to be disruptive to energy grids. On the other hand, a few of the smaller clusters around large wildfire footprints don’t rank highly in the risk calculation, meaning they had smaller outage or network betweenness impact. With this figure, we can visually see the risk score not ranking solely off of proximity to wildfire or size of energy infrastructure, which displays the pipeline is properly accounting for the various risk models we have chosen to incorporate.

5 Discussions

Our analysis combined direct exposure metrics (burned acres, severity, proximity to fire perimeters) with regional hazard indicators derived from clustering (DBSCAN) and density estimation (Gaussian KDE). This composite risk metric offers a multi-layered view



(a) DBSCAN clusters of wildfires



(b) Gaussian KDE hotspots of wildfires

Figure 6: Comparison of DBSCAN clustering and KDE hotspots for wildfires.

of vulnerability across substation clusters resembling key areas of energy infrastructure. The results highlight that infrastructure risk is not solely a function of past fire overlap but also has a significant spatial element as clusters located near persistent wildfire regions or within high-density wildfire areas exhibit elevated risk scores even if they have not yet experienced direct damage.

From a decision-making perspective, these findings can inform utility planning and resilience planning decisions. Policymakers and other planners can use the composite risk scores to prioritize resiliency measures in clusters with high exposure and high risk of wildfires. Emergency managers may also leverage the DBSCAN clusters to identify regions where repeated fire activity threatens critical infrastructure, while KDE hotspots provide a probabilistic surface for anticipating future ignitions. While individually performing these analyses can be key, they may not be interpretable to decision makers, highlighting the importance of a simple risk score that gives a reference as to the degree to which a certain cluster of energy infrastructure is at risk.

Despite the utility of this scoring metric, there are important ethical considerations when using this model. It must be noted that this provides an indication of relative vulnerability and are not guarantees of failure. Users must be cautious to avoid over-relying on these outputs to prevent an inequitable allocation of resources to communities across the state. There must be some assurance that communities outside the identified clusters still receive adequate support, despite being considered lower risk. Policymakers must be transparent with how these metrics are calculated, and ensure that the implemented resilience strategies do not disproportionately burden vulnerable populations in California.

In summary, our composite risk framework demonstrates how integrating clustering analysis with direct exposure metrics yields actionable insights for infrastructure resilience against wildfires. The approach is not a replacement for expert opinion, but instead, is a tool that complements existing planning strategy to guide planning, allowing for more transparent decision-making as wildfires become more prevalent.

6 Limitations

Our current analysis relies on a static geographic representation of the grid, which does not incorporate real time flow of energy data that describes direction and magnitude. In real environments substations actively receive and deliver energy over time with demand and outages. Our betweenness metric assumes the paths are used equally, so idle substations may have overemphasized importance values.

Additionally, this work could be improved through further collaboration with environmental scientists and urban planners, particularly with regards to energy, to determine appropriate weighting of components in our risk metric.

Another limitation to consider is the lack of data around average loads of particular substations. When computing a substation's outage impact, we would ideally value one with higher typical loads more than one with less.

The analysis performed in this paper is also limited by spatial simplification, as these don't fully take into account metrics such as terrain, wind, and fuel (vegetation) [5] which would impact

wildfire risk assessments. As an adjacent idea, since we currently represent substations as points we might also miss the idea of longer transmission lines crossing through higher risk areas.

Our results should be considered as potential candidates of risk instead of definitive weaknesses in these real world systems.

7 Conclusion

To conclude, we presented a composite risk framework to assess the impact of wildfires on California's energy infrastructure by integrating a wide array of direct exposure metrics, regional impact, and infrastructure network systems. This approach highlighted vulnerable clusters of substations and provided a data-driven metric to support resilience planning for future energy infrastructure construction. This framework is intended to serve as a support tool for decision-making, rather than a firm predictor. Future work can be extend this analysis with finer-scale data, such as real-time load information and more connection data between plants and substations, ensuring more accurate scoring metrics for improved planning.

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9 Author Contributions

Both authors contributed equally for this work, with Aziz responsible for wildfire network analysis as well as the clustering for this layer, and Nicholas responsible for general energy infrastructure analysis and the associated clustering. The testing and development of the risk scoring was performed by both authors, along with the creation of this report.

10 Data and Code Availability

All code developed for this report is available at: <https://github.com/azizs1/wildfires-energy-infra-impact>.

The data used in this paper was obtained from the following sources:

- MTBS Burned Area Boundaries Dataset: Monitoring Trends in Burn Severity (MTBS), USGS/US Forest Service. <https://mtbs.gov/direct-download>
- HIFLD US Electric Power Transmission Lines Dataset: Homeland Infrastructure Foundation-Level Data (HIFLD), U.S. Department of Homeland Security. <https://www.kaggle.com/datasets/behroozsohrabi/us-electric-power-transmission-lines>
- HIFLD US Electric Power Plants Dataset: Homeland Infrastructure Foundation-Level Data (HIFLD), U.S. Department of Homeland Security. <https://www.kaggle.com/datasets/behroozsohrabi/us-electric-power-plants>
- HIFLD US Electric Power Transmission Substations Dataset: Homeland Infrastructure Foundation-Level Data (HIFLD), U.S. Department of Homeland Security. <https://www.kaggle.com/datasets/behroozsohrabi/us-electric-power-transmission-substations>

[com/datasets/behroozsohrabi/us-electric-power-transmission-substations](https://www.kaggle.com/datasets/behroozsohrabi/us-electric-power-transmission-substations)

All datasets are publicly accessible and were used in accordance with their respective licensing and usage policies.

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